ECE 310 Fall 2026
Lecture 3: Discrete-time systems

Corey Snyder

University of Illinois Urbana-Champaign

August 28th, 2026



Learning Objectives

- Define what represents a discrete-time system.
- State the conditions for systems to be linear, time-invariant, and causal.
- State the definition of a system being bounded-input bounded-output (BIBO) stable.
- Prove whether a given discrete-time system is linear, time-invariant, causal, or stable.



Discrete-time Systems

- A *discrete-time system* is any computational process that maps a discrete-time input, $x[n]$, to a discrete-time output, $y[n]$.
- Let T denote an abstract system or *operator* that maps $x[n]$ to $y[n]$.

$$x[n] \overset{T}{\mapsto} y[n]$$
$$y[n] = T(x[n])$$



Discrete-time Systems

- A *discrete-time system* is any computational process that maps a discrete-time input, $x[n]$, to a discrete-time output, $y[n]$.
- Let T denote an abstract system or *operator* that maps $x[n]$ to $y[n]$.

$$x[n] \overset{T}{\mapsto} y[n]$$
$$y[n] = T(x[n])$$

- Examples:

$$y[n] = x[n] - x[n-1]$$
$$y[n] = \text{median}\{x[n], x[n-1], x[n-2]\}$$
$$y[n] = \text{SnapchatFilter}(x[n])$$



Linearity

A discrete-time system T is *linear* if and only if it satisfies the following two conditions:

$$T(ax[n]) = aT(x[n]),\ a \in \mathbb{R} \qquad \text{(Homogeneity)}$$

$$T(x_1[n] + x_2[n]) = T(x_1[n]) + T(x_2[n]) \qquad \text{(Additivity)}$$



Linearity

Linearity

A discrete-time system T is *linear* if and only if it satisfies the following two conditions:

$$T(ax[n]) = aT(x[n]),\ a \in \mathbb{R} \qquad \text{(Homogeneity)}$$
$$T(x_1[n] + x_2[n]) = T(x_1[n]) + T(x_2[n]) \qquad \text{(Additivity)}$$

which is equivalent to the following single condition:

$$T(ax_1[n] + bx_2[n]) = aT(x_1[n]) + bT(x_2[n]),\ a, b \in \mathbb{R}\ \text{(Superposition)}.$$



menti.com: 2105 8661

Linearity

A discrete-time system T is *linear* if and only if it satisfies the following two conditions:

$$T(ax[n]) = aT(x[n]),\ a \in \mathbb{R} \qquad \text{(Homogeneity)} \quad (1)$$

$$T(x_1[n] + x_2[n]) = T(x_1[n]) + T(x_2[n]) \qquad \text{(Additivity)} \quad (2)$$

which is equivalent to the following single condition:

$$T(ax_1[n] + bx_2[n]) = aT(x_1[n]) + bT(x_2[n]),\ a, b \in \mathbb{R}\ \text{(Superposition)}. \quad (3)$$

In summary, we can prove a system is linear by showing it satisfies homogeneity **and** additivity; **or**, that the system satisfies superposition.

Try guessing and checking
- Guess non-linear → Try to disprove (1) or (2)
- Guess linear → Try to prove (3)

Linearity Practice

Ex. 1: $y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$

Ex. 2: $y[n] = x[n] + 1$

Guess linear and check superposition

$$T(aX_1[n] + bX_2[n]) \stackrel{?}{=} aT(X_1[n]) + bT(X_2[n])$$

Let $z[n] = aX_1[n] + bX_2[n]$,

$$T(z[n]) = \frac{1}{2}z[n] + \frac{1}{2}z[n-1]$$

$$= \frac{1}{2}(aX_1[n] + bX_2[n]) + \frac{1}{2}(aX_1[n-1] + bX_2[n-1])$$

$$= a\underbrace{\left(\frac{1}{2}X_1[n] + \frac{1}{2}X_1[n-1]\right)}_{T(X_1[n])} + b\underbrace{\left(\frac{1}{2}X_2[n] + \frac{1}{2}X_2[n-1]\right)}_{T(X_2[n])}$$

$$= aT(X_1[n]) + bT(X_2[n]) \checkmark \text{ (Satisfies superposition and is linear)}$$

Linearity Practice

Ex. 1: $y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$

Ex. 2: $y[n] = x[n] + 1$

Guess non-linear:

Check homogeneity: $T(ax[n]) \stackrel{?}{=} aT(x[n])$

- $T(ax[n]) = ax[n] + 1$

  Let $v[n] = ax[n] \rightarrow T(v[n]) = v[n] + 1 = ax[n] + 1$

- $aT(x[n]) = a(x[n] + 1) = ax[n] + a \rightarrow$ Not the same!

  $\downarrow$

  Non-Linear

Extra Linearity Practice

Ex. 1: $y[n] = |x[n]|$

* $T(a x[n]) \stackrel{?}{=} a T(x[n])$
* $T(a x[n]) = |a x[n]|$
* $a T(x[n]) = a|x[n]|$

$\Rightarrow$ Non-linear ✗

Ex. 2: $y[n] = n(x[n] - x[n-1])$

* $T(a x[n]) \stackrel{?}{=} a T(x[n])$
* $T(a x[n]) = n(a x[n] - a x[n-1])$
  $= a[n(x[n] - x[n-1])]$
  $= a T(x[n])$

Homogeneity ✓

* $T(x_1 + x_2) \stackrel{?}{=} T(x_1) + T(x_2)$
* $T(x_1 + x_2) = n[(x_1[n] + x_2[n]) - (x_1[n-1] + x_2[n-1])]$
  $= n(x_1[n] - x_1[n-1]) + n(x_2[n] - x_2[n-1])$
  $= T(x_1[n]) + T(x_2[n])$

Additivity ✓

→ Linear

Ex. 3: $y[n] = e^{x[n]}$

* $T(a x[n]) \stackrel{?}{=} a T(x[n])$
* $T(a x[n]) = e^{a x[n]}$
* $a T(x[n]) = a e^{x[n]}$

$\Rightarrow$ Non-linear ✗

Ex. 4: $y[n] = x[n]x[0]$

* $T(a x[n]) \stackrel{?}{=} a T(x[n])$
* $T(a x[n]) = (a x[n])(a x[0])$
  $= a^2 x[n] x[0]$
* $a T(x[n]) = a x[n] x[0]$

$\Rightarrow$ Non-linear ✗

Time-invariance

Time-invariance

A discrete-time system T is *time-invariant* or *shift-invariant* if and only if

$$y[n - n_0] = T(x[n - n_0]),\ n \in \mathbb{Z}$$

for input $x[n]$ and output $y[n]$. In different notation, this may be written as

$$T(x)[n - n_0] = T(x[n - n_0]),\ n \in \mathbb{Z}.$$



Time-invariance

A discrete-time system T is *time-invariant* or *shift-invariant* if and only if

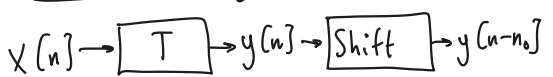
$$y[n - n_0] = T(x[n - n_0]),\ n \in \mathbb{Z}$$

for input $x[n]$ and output $y[n]$. In different notation, this may be written as

$$T(x)[n - n_0] = T(x[n - n_0]),\ n \in \mathbb{Z}.$$

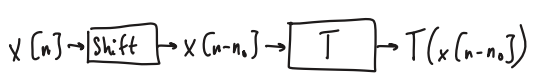
In words, this means that shifting the input $x[n]$ by n_0 samples leads to a corresponding shift of n_0 samples in the output.

Left side: $y[n-n_0]$ or $T(x)[n-n_0]$



* Replace every n with $n-n_0$ for the output

Right side: $T(x[n-n_0])$



Time-invariance Practice

Ex. 1: $y[n] = x[n] - x[n-1]$

Ex. 2: $y[n] = x[2n]$

Example 1:

$y[n-n_0] \stackrel{?}{=} T(x[n-n_0])$

- LS: $y[n-n_0] = x[(n-n_0)] - x[(n-n_0)-1]$
- RS: $T(x[n-n_0]) = T(v[n])$ ← let $v[n] = x[n-n_0]$

  $= v[n] - v[n-1]$

  $= x[n-n_0] - x[n-n_0-1]$

Two sides are equivalent, so the system is time-invariant!

Time-invariance Practice

Ex. 1: $y[n] = x[n] - x[n-1]$

Example 2:

$y[n-n_0] = T(x[n-n_0])$

Left side: $y[n-n_0] = x[2(n-n_0)] = x[2n-2n_0]$

Right side: $T(x[n-n_0]) = T(v[n])$

(Let $v[n] = x[n-n_0]$) $= v[2n]$

$= x[2n-n_0]$

→ Two sides are different so time-varying! (Not time-invariant)

Ex. 2: $y[n] = x[2n]$

Ex: $x[n] = \{2, 4, 6, 1, 3, 5\}$ (2 is the 0th sample)

Let $n_0 = 1$

- $T(x[n]) = x[2n] = \{2, 6, 3\}$
- $x[n-1] = \{0, 2, 4, 6, 1, 3, 5\}$ (0 is the 0th sample)
- LS: $y[n-1] = \{0, 2, 6, 3\}$ (0 is the 0th sample)
- RS: $T(x[n-1]) = \{0, 4, 1, 5\}$ (0 is the 0th sample)

$T(x[n]) \neq y[n]$

This is a counter-example and thus this system is time-varying.

Extra Time-invariance Practice

Ex. 1: $y[n] = x[|n|]$

apply abs. value to time index

* $y[n-n_0] = x[|(n-n_0)|]$
* $T(x[n-n_0]) = x[|n|-n_0]$ ≠ $y[n-n_0]$

$\Longrightarrow$ Time-varying

Ex. 2: $y[n] = n(x[n] - x[n-1])$

* $y[n-n_0] = (n-n_0)(x[n-n_0] - x[(n-n_0)-1])$
* $T(x[n-n_0]) = n(x[n-n_0] - x[n-n_0-1])$

only input is shifted by applying system!

$\Longrightarrow$ Time-varying

replace every n with $(n-n_0)$

Ex. 3: $y[n] = e^{x[n]}$

* $y[n-n_0] = e^{x[n-n_0]}$
* $T(x[n-n_0]) = T(z[n])$ $\quad (z[n] = x[n-n_0])$
  $= e^{z[n]}$
  $= e^{x[n-n_0]}$

$\Longrightarrow$ Time-invariant

Ex. 4: $y[n] = x[n]x[0]$

original $x[0]$ is locked in before shifting the output!

* $y[n-n_0] = x[n-n_0]x[0]$
* $T(x[n-n_0]) = x[n-n_0]x[0-n_0]$

Shifting by n_0 changes the sample at $x[0]$!

$\Longrightarrow$ Time-varying

Causality

Causality

A discrete-time system is **causal** if the system output only depends on **present** or **previous** input and output samples. In other words, the system does not depend on future samples.

Ex. 1: $y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$

$y[n]$ depends on: $x[n]$ (present)

$x[n-1]$ (past)

$x[n-2]$ (past)

↓

Only present + past

→ Causal

Ex. 2: $y[n] = x[|n|]$

$y[n]$ depends on $x[|n|]$

$y[-2] = x[|-2|]$

$= x[2]$

↓

In the future

↓

Non-causal!

Extra Causality Practice

Ex. 1: $y[n] = x[|n| - n]$

* $y[-1] = x[|-1| - (-1)]$
  $= x[2]$

$\Rightarrow$ Non-causal

Ex. 3: $y[n] = e^{x[n]}$

* $y[n]$ at index n depends on input $x[n]$ (present).

$\Rightarrow$ Causal

Ex. 2: $y[n] = (n+1)(x[n] - x[n-1])$

* $y[n]$ at index n depends on inputs $x[n]$ (present) and $x[n-1]$ (past)

$\Rightarrow$ Causal

* Note: the $(n+1)$ does not affect which input samples we use!

Ex. 4: $y[n] = x[n]x[0]$

* $y[-1] = x[-1]x[0]$ ← future input

$\Rightarrow$ Non-causal

BIBO Stability

BIBO stability

- A discrete-time system T is *bounded-input bounded-output stable* (BIBO stable) if and only if for any $x[n]$ such that $|x[n]| < \beta$ for all n, $|T(x[n])| < \alpha$ for all n where $0 \leq \alpha < \infty,\ 0 \leq \beta < \infty$.
- In words, a system is BIBO stable if and only if the system produces a bounded output signal for any bounded input signal. A signal is bounded if it takes a finite value for all indices $n \in \mathbb{Z}$.

Ex. 1: $y[n] = \sin^2(x[n])$

- $|x[n]| \leq B$ (bounded input)

$\rightarrow 0 \leq \sin^2(x[n]) \leq 1$

$\downarrow$

$0 \leq |y[n]| \leq 1 < \infty \rightarrow$ BIBO Stable

Ex. 2: $y[n] = \ln(|x[n]|)$

$|x[n]| \leq B$

$\rightarrow |y[n]| \leq \ln(B)$

→ However, $x[n]$ could be 0 and $\ln(0) \rightarrow -\infty$

→ Not stable

Extra Stability Practice

Ex. 1: $y[n] = \frac{x[n]}{|n|+1}$

$$|y[n]| = \frac{|x[n]|}{||n|+1|}$$

$$\leq \frac{\beta}{|n|+1}$$

$$\leq \beta$$

$$< \infty$$

BIBO stable

Ex. 2: $y[n] = n(x[n] - x[n-1])$

$$|y[n]| = |n(x[n] - x[n-1])|$$

$$= |n|(|x[n] - x[n-1]|)$$

$$\leq |n|\,|\beta| \to \infty \text{ as } n \to \infty$$

$\Longrightarrow$ Unstable

Ex. 3: $y[n] = e^{x[n]}$

$$|y[n]| = |e^{x[n]}|$$

$$\leq e^{|x[n]|}$$

$$\leq e^{\beta}$$

$$< \infty$$

BIBO stable

Ex. 4: $y[n] = x[n]x[0]$

$$|y[n]| = |x[n]x[0]|$$

$$= |x[n]| \cdot |x[0]|$$

$$\leq \beta^2$$

$$< \infty$$

BIBO stable